

# The trajectories of neighborhood change in historic districts within a growing context

Sunho Choi <sup>\*</sup>

## ARTICLE INFO

### Keywords:

Historic districts  
Neighborhood change  
Socioeconomic transitions  
Neighborhood typology  
Urban growth

## ABSTRACT

Local historic districts to preserve the character of built environments in neighborhoods function as certifying their historical significance and controlling their transformations. These functions can influence the physical fabric of neighborhoods, and their impacts may extend to socioeconomic characteristics. This study investigates the relationship between the designation of local historic districts and neighborhood change within the context of growth by exploring socioeconomic transitions in eight Western cities. Based on six neighborhood types identified by k-means cluster analysis with tract-level data between 1970 and 2020, I examine neighborhood trajectories of local historic districts compared to undesignated neighborhoods and the distinct trends in neighborhood change by the phases of district designation. This research finds that neighborhoods with local historic districts were more likely to experience upgrading movements than undesignated neighborhoods, especially prevalent during and after designation compared to the pre-designation period. The trajectories of neighborhood change in local historic districts also varied across the neighborhood types, implying a greater possibility of gentrification and displacement in unstable lower-income neighborhoods. These results highlight the potential influence of local historic districts as a catalyst for neighborhood change, requesting an integrated approach to mitigate the socioeconomic impacts of district designation for equitable and inclusive preservation practices.

## 1. Intro

The primary focus of historic districts is to preserve the distinctive character of built environments in neighborhoods, which is rooted in several precedents aimed at managing the quality of built environments within larger areas rather than individual structures (Birch & Roby, 1984; Hamer, 1998). With this goal of neighborhood preservation, the designation of historic districts in many American cities certifies the historical significance of neighborhoods and controls their transformations through regulations. These functions of historic districts over large areas may greatly influence the physical fabric of neighborhoods, and their impacts may extend to socioeconomic characteristics. A substantial body of literature has examined these neighborhood effects of historic districts, mostly focusing on property values (Been et al., 2016; Coulson & Lahr, 2005; Heintzelman & Altieri, 2013; Leichenko et al., 2001; Noonan, 2007; Noonan & Krupka, 2011; Oba & Noonan, 2020; Zahirovic-Herbert & Chatterjee, 2012). In line with these studies, some scholars cautioned that the price premium attributed to district designation may serve as a trigger of gentrification and displacement (Fein, 1985; Listokin et al., 1998; Smith, 1998; Werwath, 1998; Zukin, 1987).

However, empirical research exploring the relationship between historic districts and neighborhood socioeconomic change remains in the nascent stage, so such claims may not be valid in all cases (Coulson & Leichenko, 2004; Kinahan & Ruther, 2020; McCabe & Ellen, 2016).

Neighborhoods in rapidly growing cities demonstrate a broad range of trajectories in their socioeconomic transition, influenced by the increasing development pressure from the continuous influx of population (Delmelle, 2015; Foote & Walter, 2017). Within this context of growth, the primary role of historic preservation is often regarded as conserving neighborhood characteristics to protect old neighborhoods and their cultural heritage from the threat of demolition and redevelopment. Otherwise, preservation programs and policies may also be employed within redevelopment systems as part of culture-driven strategies for neighborhood revitalization (Kinahan & Mawhorter, 2021). Given the complexity of neighborhood change and the different functions of historic preservation within the context of growth, the trajectories of neighborhood socioeconomic change influenced by the designation of historic districts may be heterogeneous, not just gentrification. Therefore, exploring the patterns of socioeconomic transitions in local historic districts provides empirical evidence that highlights

<sup>\*</sup> Department of Urban Design and Planning, University of Washington, Seattle, WA 98105, USA.

E-mail address: [sunhoch@uw.edu](mailto:sunhoch@uw.edu).

<https://doi.org/10.1016/j.apgeog.2024.103478>

Received 4 July 2024; Received in revised form 31 October 2024; Accepted 25 November 2024

Available online 4 December 2024

0143-6228/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

their roles as a neighborhood planning tool within the context of growth, which also contributes to the recent discourse on equitable and inclusive historic preservation (Arlotta & Avrami, 2020; Cheong, 2021; Gao & Dublin, 2020; Minner, 2016; Wells, 2021).

This study examines the relationship between the designation of local historic districts and neighborhood socioeconomic change using a multivariate classification technique on longitudinal tract-level data from 1970 to 2020. In order to investigate this relationship within the context of growth, this research addresses eight cities in the Western region: Long Beach, Los Angeles, Oakland, San Diego, and San Francisco, California; Denver, Colorado; Portland, Oregon; and Seattle, Washington. Even as these cities experienced massive population growth and dramatic spatial transformations, they began designating local historic districts in the 1970s to preserve their cultural heritage and stabilize neighborhoods (Morley, 2006). This study explored trajectories of neighborhood transitions in these cities using a typology identified through k-means cluster analysis. The following research questions are addressed. First, what are the transition patterns and neighborhood characteristics of local historic districts compared to undesignated neighborhoods? Second, how does the association of locally designated historic districts with neighborhood stability vary across different neighborhood types? Third, how do the patterns of neighborhood change differ by the phases of district designation: before, during, and after designation?

## 2. Neighborhood change and historic districts

Since the emergence of historic preservation as an alternative to urban renewal and following the National Historic Preservation Act of 1966, preservation policies and programs have been widely practiced across the United States. As the experience of preservation practices has grown in urban settings, there have been increasing questions about socioeconomic impacts. A body of preservation literature has examined only the economic contributions of historic preservation — such as the impacts on regional economy, heritage tourism, property values, neighborhood revitalization, and community development (Listokin et al., 1998; Mason, 2005; Ryberg-Webster & Kinahan, 2014; Rypkema et al., 2013). Although possible connections with gentrification and displacement began receiving attention in the 1980s (Fein, 1985; Zukin, 1987), the discourse regarding the social influence of historic preservation, especially through the lens of equity and inclusion, has only recently emerged with a growing call for social justice in the field (Arlotta & Avrami, 2020; Cheong, 2021; Minner, 2016; Wells, 2021).

In this discussion of the impacts of preservation policies, historic districts are considered particularly significant for their broad influences at the neighborhood scale. Their basic principle of preserving the character of the built environment in neighborhoods was developed through several precedents to manage a larger area over an individual building, such as Colonial Williamsburg's museum village, the Charleston Old and Historic District, and the College Hill plan in Providence (Birch & Roby, 1984; Hamer, 1998; Minner, 2016). Based on this primary focus of historic districts on neighborhood preservation, the designation of historic districts officially certifies the historical significance of the built environment in the designated area. Moreover, the designation of local historic districts typically regulates external changes within their boundary through the design review process by a historic preservation commission or board. The regulatory controls over historic properties within local historic districts result in decreasing demolition rates (Kinahan & Mawhorter, 2021) and new housing constructions (Been et al., 2016). Also, these controls and the certification function of historic districts can catalyze various trajectories of socioeconomic changes in neighborhood characteristics as well as spatial transformations.

Empirical studies examining the impacts of historic districts on property values and sales prices constitute the largest part of the literature on their socioeconomic influences. The underlying assumptions of

these studies suggest that the designation of historic districts provides price premiums and ensures neighborhood stability through honorary status and regulatory controls (Heintzelman & Altieri, 2013). The designation may offer a cachet for historic properties leading to increases in their values compared to those in undesignated neighborhoods, while it can provide stability by guaranteeing minimal changes and preserving the current state in the designated area. Economic incentives such as grants and tax credits followed from historic district designation often attract interest from developers and investors (Fein, 1985). Many studies support the assumption by providing empirical evidence of the positive influences of district designation on property values and sales prices (Been et al., 2016; Coulson & Lahr, 2005; Leichenko et al., 2001; Zahirovic-Herbert & Chatterjee, 2012), whereas a few other studies using longitudinal datasets and more developed statistical models reveal the complexity of the relationship (Heintzelman & Altieri, 2013; Noonan, 2007; Noonan & Krupka, 2011). In particular, several recent studies acknowledge the endogeneity issues in the designation of historic districts, where designation decisions may be influenced by unobserved neighborhood factors such as unique property characteristics, local market trends, and neighborhood collective actions (Ahlfeldt et al., 2017; Been et al., 2016; Noonan & Krupka, 2011; Zhou, 2021). These studies utilize more sophisticated statistical approaches to separate the impacts of historic designation on property values from the endogenous factors.

The price premium associated with district designation often encourages investment in historic properties and contributes to neighborhood revitalization. However, these economic benefits are often recognized as an accelerator of gentrification by imposing considerable financial burdens on both owners and renters, potentially leading to the displacement of those who cannot afford to pay increased property taxes and rents (Fein, 1985; Listokin et al., 1998; Smith, 1998; Werwath, 1998; Zukin, 1987). In particular, lower-income populations may face a higher threat of gentrification and displacement because the increase in property values attributed to historic district designation could be greater for lower-priced properties compared to higher-priced properties, potentially leading to increased housing costs for these households (Oba & Noonan, 2020; Zahirovic-Herbert & Chatterjee, 2012). Additionally, neighborhood change in historic districts may be associated with the consumption patterns of specific population groups who value various cultural amenities and social experiences (Allison & Peters, 2011; Gotham, 2005; Zukin, 1987). The authentic and unique environments of historic districts commonly attract moderate- or upper-income residents and higher-educated people.

Although a large body of preservation literature concentrates on the association of district designation with price premiums and gentrification, there are several different trajectories of neighborhood change related to historic districts. Neighborhood stability has been one of the shared goals between preservationists and planners since the late 1960s when the direction of federal urban policy transitioned from demolition-based urban renewal projects to preservation-led community stabilization programs (Birch & Roby, 1984). Historic districts have been used as a neighborhood planning tool by neighborhood residents and community activists to achieve the neighborhood stabilization goal. For example, many property owners in New Haven's City Point local historic district regarded the primary benefit of district designation as stabilizing their neighborhood against higher crime and lower owner-occupancy rates (Heuer, 2007). Also, many community organizations working for neighborhood stabilization continue to use incentives for preservation activities eligible within historic districts to create and provide affordable housing for those who face a higher risk of displacement (Cohen, 1998; Passell, 2021; Ryberg-Webster, 2019).

While a body of preservation literature explores the relationship between historic district designation and neighborhood socioeconomic change, qualitative studies have typically focused on specific cases. Quantitative studies have also broadened their geographic scope into larger areas, but their focus has been mostly limited to the impacts on

**Table 1**  
Summary of the studied cities.

| Cities            | Population |           |         | Average Older Buildings%* |      |         |
|-------------------|------------|-----------|---------|---------------------------|------|---------|
|                   | 1970       | 2020      | Change% | 1970                      | 2020 | Change% |
| Long Beach, CA    | 358,633    | 466,742   | 30.1    | 29.1                      | 91.5 | 214.4   |
| Los Angeles, CA   | 2,816,061  | 3,898,747 | 38.4    | 30.5                      | 87.2 | 185.9   |
| Oakland, CA       | 361,561    | 440,646   | 21.9    | 52.6                      | 86.8 | 65.0    |
| San Diego, CA     | 696,769    | 1,386,932 | 99.1    | 16.2                      | 82.8 | 411.1   |
| San Francisco, CA | 715,674    | 873,965   | 22.1    | 64.1                      | 89.1 | 39.0    |
| Denver, CO        | 514,678    | 715,522   | 39.0    | 32.8                      | 81.1 | 147.3   |
| Portland, OR      | 382,619    | 652,503   | 70.5    | 46.4                      | 78.1 | 68.3    |
| Seattle, WA       | 530,831    | 737,015   | 38.8    | 46.7                      | 72.1 | 54.4    |

\*The “average older buildings%” refers to the average proportion of buildings over 30 years old in each census tract within each city, calculated as the unweighted mean across tracts.

property values without other socioeconomic variables. Therefore, significant research gaps remain in preservation research, as empirical studies using large longitudinal datasets to explore multivariate socioeconomic changes in designated neighborhoods are still in the early stages. One of the initial studies exploring a case of Fort Worth, Texas, examines this relationship by setting the presence and quantity of designated properties as the sole independent variable,<sup>1</sup> yet it did not find significant changes in neighborhood demographic and housing features (Coulson & Leichenko, 2004). However, some recent research provides empirical evidence that suggests the possibility of gentrification and displacement attributed to district designation. A study investigating neighborhood change in local historic districts in comparison to other neighborhoods in New York City found significant increases in socioeconomic status and some degree of changes in the racial composition following district designation (McCabe & Ellen, 2016). Further analysis extending over 46 U.S. cities also demonstrates significant decreases in African-American and Hispanic populations after the creation of local historic districts and National Register historic districts (Kinahan & Ruther, 2020). Moreover, a recent study examining property transactions within Denver’s historic districts suggests that district designations may exacerbate residential segregation by promoting transaction flow primarily among White sellers and White buyers (Bologna Pavlik & Zhou, 2023). While these recent studies investigated changes in specific neighborhood variables after designating historic districts, the question of the comprehensive relationship between historic district designation and multiple neighborhood characteristics still remains. In the following section, I introduce a multivariate cluster approach based on a set of neighborhood demographic, socioeconomic, and housing variables to examine the trajectories of neighborhood change in local historic districts within the context of growth.

<sup>1</sup> Although Coulson and Leichenko (2004) did not use district designation as an independent variable, using the number of designated properties implicitly examines the relationship between historic districts and neighborhood change. Tracts with historic districts typically have a much larger number of designated properties than other tracts with fewer landmarks.

### 3. Methods

#### 3.1. Selected cities

In order to explore differences in neighborhood socioeconomic change between local historic districts and undesignated neighborhoods in the context of growth, this study selected eight large cities within the Western region: Long Beach, Los Angeles, Oakland, San Diego, and San Francisco, California; Denver, Colorado; Portland, Oregon; and Seattle, Washington (Table 1). The selected cities were the most populous cities in this region, exceeding 300,000 in 1970,<sup>2</sup> and their populations have drastically increased over the last 50 years. Within the context of growth, the notable patterns of spatial transformations to satisfy increasing housing demands include the redevelopment of older neighborhoods as well as sprawl development in the fringe areas (Delmelle, 2015). Although the construction of new structures after the demolition of older buildings is the most common redevelopment cycle, historic preservation through rehabilitation and adaptive reuse is another option that contributes to neighborhood revitalization (Kinahan & Mawhorter, 2021). These different approaches for managing spatial transformations in older neighborhoods lead to a large degree of variability in their socioeconomic change within the context of growth (Delmelle, 2015; Foote & Walter, 2017), suggesting the possibility of more subtle distinctions in the trajectories of neighborhood transitions comparing local historic districts and undesignated neighborhoods.

In the Western region, the designation of local historic districts has been crucial for neighborhood stabilization amidst the continuous waves of newcomers. In the 1960s, these cities began to adopt municipal preservation ordinances. Beginning in the 1970s, they started designating local historic districts to preserve cultural heritage and historic places in the designated areas, with the ultimate goal of stabilizing neighborhoods and enhancing a sense of community (Morley, 2006). In the 1990s and 2000s, the establishment of new local historic districts

<sup>2</sup> Three additional cities in this region — Honolulu, Hawaii; San Jose, California; and Phoenix, Arizona — satisfied the population criterion, but they were excluded due to data limitations impacting the analysis. First, Honolulu does not address local historic districts in its municipal preservation ordinance. Second, San Jose, despite having six local historic districts, has only three local historic districts of adequate size for analysis. These three local historic districts occupy less than 2% of the total tracts in the city, which is insufficient for reliable analysis. Lastly, in Phoenix, only eight census tracts out of the city’s 384 tracts are classified as older neighborhoods appropriate for the study. Only six of the 36 total local historic districts are located in these older neighborhoods. Given the small number of suitable tracts, reliable analysis is not feasible.

**Table 2**

Summary of local historic districts (excluding non-residential areas) in the studied cities.

| Cities        | First Designation | 1970–1979 | 1980–1989 | 1990–1999 | 2000–2009 | 2010–2019 |
|---------------|-------------------|-----------|-----------|-----------|-----------|-----------|
| Long Beach    | 1982              | –         | 3         | 9         | 5         | 1         |
| Los Angeles   | 1983              | –         | 3         | 4         | 17        | 11        |
| Oakland       | 1975              | 2         | 2         | 1         | 3         | –         |
| San Diego     | 1970              | 3         | 4         | –         | 6         | 5         |
| San Francisco | 1972              | 1         | 6         | 3         | 1         | 3         |
| Denver        | 1973              | 9         | 4         | 13        | 13        | 8         |
| Portland      | 1975              | 2         | 3         | 9         | 2         | 1         |
| Seattle       | 1970              | 5         | 1         | –         | –         | –         |
| <b>Total</b>  | –                 | <b>22</b> | <b>26</b> | <b>39</b> | <b>47</b> | <b>29</b> |

increased significantly in most of these cities<sup>3</sup> with the aging of built environments within these cities (Table 2). The average proportion of buildings constructed more than 30 years ago in a census tract has grown from 37% in 1970 to 57% in 1990, and to 83% in 2020. As these older built fabrics still pose a question about how to manage them in consideration of social and economic tensions, some of the study cities develop new neighborhood conservation tools to protect historic fabrics and neighborhood characteristics within the rapid urban transformations (Buckley & Graves, 2016; Idziorek & Chalana, 2019; Napoli et al., 2019).

### 3.2. Longitudinal neighborhood data

For the longitudinal analysis of neighborhood change in the eight Western cities from 1970 to 2020, this study uses the Neighborhood Change Database (NCDB) from GeoLytics (2024). The NCDB standardizes the Decennial Census from 1970 to 2020, as well as the American Community Survey data from 2006 to 2010 and 2016–2020, by adjusting different tract boundaries in each decade to match those of the 2020 census tracts.<sup>4</sup> The standardization allows for investigating trajectories of neighborhood transformations over the study period within the same tract boundaries. Therefore, this study used the 2020 census tracts as a proxy for neighborhoods to examine their transitions in accordance with prior research on neighborhood typologies and socioeconomic restructuring (Delmelle, 2015; Foote & Walter, 2017; Mikelbank, 2011; Owens, 2012; Wei & Knox, 2014).

Based on the previous studies and data availability from the NCDB, this study selected 20 neighborhood variables categorized into demographic (race/ethnicity, immigrant status, age, and family structure), socioeconomic (educational attainments, occupations, unemployment, and household income), and housing characteristics (vacancies, homeownership, and units in structure) (Table 3). Census tracts with insufficient data or errors on any of the variables during the study period were excluded from the cluster analysis. Also, I removed tracts with exceptionally low populations (less than 100), low households (less than

**Table 3**

A list of variables for cluster analysis.

| Variables                            | Descriptions                                                                      |
|--------------------------------------|-----------------------------------------------------------------------------------|
| <i>Demographic</i>                   |                                                                                   |
| White%                               | Prop. of White population                                                         |
| Black%                               | Prop. of Black/African American population                                        |
| Hispanic%                            | Prop. of Hispanic/Latino population                                               |
| Immigrant%                           | Prop. of population who are foreign born                                          |
| Child%                               | Prop. of persons 0–17 years old                                                   |
| Adult%                               | Prop. of persons 18–64 years old                                                  |
| Aging%                               | Prop. of persons 65+ years old                                                    |
| Nonfamily%                           | Prop. of nonfamily households                                                     |
| <i>Socioeconomic Characteristics</i> |                                                                                   |
| High School%                         | Prop. of persons 25+ years old who have completed high school but no college      |
| College%                             | Prop. of persons 25+ years old who have a bachelor's or graduate degree           |
| Professional%                        | Prop. of persons 16+ years old employed in professional and technical occupations |
| Manual%                              | Prop. of persons 16+ years old employed as manual laborers*                       |
| Unemployment %                       | Prop. of persons 16+ years old who are unemployed                                 |
| Poverty%                             | Prop. of total persons below the poverty level last year                          |
| HH Income                            | Average household income last year                                                |
| <i>Housing Characteristics</i>       |                                                                                   |
| Vacant%                              | Prop. of vacant housing units                                                     |
| RenterOcc%                           | Prop. of renter-occupied housing units                                            |
| Singleunit%                          | Prop. of housing units consisting of 1 unit, detached                             |

\* Manual laborers include precision production/craft/repair workers, operators, assemblers, transportation/material moving workers, and nonfarm laborers.

10), or low average household income (less than \$1000) to limit the influences of outliers on the cluster centroids in classifying neighborhood typologies.<sup>5</sup> In total, 13,800 census tracts in the Western region over six census years were selected for the analysis. Lastly, a city-relative z-score for each variable compared to the other tracts in the same census year was calculated to standardize variables with different scales, magnitudes, and years.

### 3.3. Cluster analysis for neighborhood typology

Cluster analysis is a statistical technique used to classify multivariate data by grouping observations into several clusters based on their similarity (for an overview of this method, see Garfinkel-Castro et al., 2020). This exploratory data approach typically relies on calculating the dissimilarity or distance between each observation to maximize intra-cluster similarity and inter-cluster dissimilarity. As cluster classifications rely on similarity across multiple variables, this method is well-suited for this research to classify neighborhood typologies using

<sup>3</sup> Seattle has not designated any local historic district in residential neighborhoods since the designation of the Harvard-Belmont Landmark District in 1980. Since then, the only two additional Landmark Districts approved were both on publicly owned former military bases: Fort Lawton Landmark District in 1988 and Sand Point Naval Air Station Landmark District in 2011.

<sup>4</sup> Geolytics, as the data provider, does not provide a detailed explanation of how they adjusted tract boundaries between different census years. However, it is likely that a similar methodology to the NCDB 2000 dataset (Geolytics, n.d.) was employed. It is assumed that they use block-level data to apply the population-weighted proportions of 1990, 2000, and 2010 tract level data to remap onto 2020 tract boundaries. They may also remap 1970 and 1980 tract data into 2020 tracts by relying on 1990 tract- and block-level data since block or block group data from 1970 to 1980 census are relatively unreliable. The final results of the tract weights would be used to normalize all five decennial census data to 2020 tract boundaries through a weighted sum for the number of population and housing units and a weighted average for medians.

<sup>5</sup> A total of 768 tracts over six census years were excluded from the analysis: 438 tracts due to insufficient data or errors, and an additional 330 tracts based on the outlier criteria.



various neighborhood characteristics (Reibel, 2011). There are two commonly used clustering techniques: k-means clustering and hierarchical clustering. The k-means cluster approach initiates by randomly placing centroids for each cluster. Then, it assigns observations to the cluster with the nearest centroid, and recalculates the centroids based on the assigned observations. This process repeats until there are no more changes in the observation assignments. In contrast, hierarchical clustering builds a hierarchy of clusters by either agglomerating smaller clusters into larger ones or dividing larger clusters into smaller ones. The agglomerative approach begins by defining each observation as an individual cluster and iteratively merging similar clusters until all observations are merged into a single cluster. Conversely, the divisive approach starts with all observations in one cluster and repeatedly splits clusters until each observation is a single cluster.

This study selected the k-means cluster approach over the hierarchical clustering technique, considering the large sample size and the flexibility of the approach as demonstrated in the previous studies (Delmelle, 2015; Foote & Walter, 2017; Wei & Knox, 2014). The k-means cluster approach is more effective and efficient in clustering large datasets. With increasing dataset sizes, the k-means approach is more likely to provide accurate results than the hierarchical technique (Rodriguez et al., 2019). Due to more complex algorithms of hierarchical clustering, the k-means cluster approach requires fewer computational resources for handling large datasets (Garfinkel-Castro et al., 2020). In addition, while the k-means clustering technique can continue to change the cluster to which each observation is assigned within the iterative process, the hierarchical approach does not allow the reassignment of observations to other clusters once they have been merged or divided into any cluster (Mikelbank, 2011; Wei & Knox, 2014). Although the need to select the number of clusters in advance of the analysis is a methodological limitation of the k-means clustering technique, this study addressed this by testing three options after specifying the optimal number of clusters using several different indices (Charrad et al., 2014).

### 3.4. Local historic districts and undesignated neighborhoods

In exploring the trajectories of neighborhood change based on historic district designation status, this study focused on local historic districts and excluded national historic districts and conservation districts. These were excluded due to different policy frameworks, which might lead to disparate effects on neighborhood dynamics. Although the studied cities use slightly different names for local historic districts, they share similar policy mechanisms in their designation criteria and regulatory controls. The designation criteria for local historic districts require the proposed areas to show historic significance and have integrity, similar to the evaluation criteria of the National Register of Historic Places (National Park Service, n.d.). Conservation districts, on the other hand, typically focus on preserving the distinctive architectural character of neighborhood built environment instead of its historic significance (Lovely, 2008; Lubens & Miller, 2002). In addition, exterior changes within local historic districts are typically reviewed by municipal preservation boards or local district commissions based on regulatory design guidelines or district ordinances, while the National Register of Historic Places does not impose any legal obligations unless properties receive financial incentives. These similarities in managing local historic districts across the cities support the assumption of similar influences on neighborhood socioeconomic transitions, even if different contexts between the cities may result in variability in outcomes.

The spatial data of local historic districts with information about their designation dates comes from data portals and planning departments of the selected cities. This study includes 163 local historic districts designated from 1970 to 2019 after excluding 16 districts that only include non-residential areas such as parks, streets, military bases, and school campuses (Table 2). Using this spatial data, I categorize each census tract into either a local historic district (LHD) tract or an undesignated (UDG) tract to compare the neighborhood transition patterns

by the designation of local historic districts. As the area of local historic districts does not correspond to census tract boundaries, this study used the five percent threshold for the categorization.<sup>6</sup> While a LHD tract has at least five percent of its area designated as local historic districts, an UDG tract has less than five percent or none of the designated area within its boundary. The use of the five percent threshold is intended to exclude tracts with too small historic district areas where the neighborhood characteristics of a tract are more likely to differ from those of historic districts within the tract. It also allows for excluding cases where a tiny portion of a historic district area is included in a tract due to geometric inconsistency between different sources of spatial data.

This study limits the sample tracts to older neighborhoods where more than half of the buildings were older than 30 years<sup>7</sup> in 1970, a criterion derived from the distribution of the older buildings ratio in LHD tracts.<sup>8</sup> This allows for the exclusion of newer neighborhoods typically ineligible for the designation of local historic districts, especially many of the UDG tracts with a large prevalence of newer buildings. As a result, 73% of the LHD tracts and 32% of the UDG tracts are selected to compare the patterns of neighborhood change between local historic districts and undesignated neighborhoods (Table 4).

## 4. Results

### 4.1. Neighborhood typology

In order to specify the best number of clusters for k-means cluster analysis, this study used the statistical package *NbClust*, which allows me to test many different indices (Charrad et al., 2014). Most of these indices suggested six neighborhood types as the appropriate number of clusters. Based on this, I tested separate k-means analyses with five, six, and seven clusters (for more details, please see Appendix A). The results from the six-group solution provided the most pertinent neighborhood typologies for analyzing the patterns of neighborhood change in local historic districts compared to undesignated neighborhoods. In this solution, each neighborhood type had distinct and interpretable characteristics and an adequately well-distributed sample size. These six neighborhood types were primarily named based on their significant

**Table 4**

The number of tracts by the designation status.

|                               | Older<br>Neighborhoods<br>(%) |        | Newer<br>Neighborhoods<br>(%) |        | Total (%) |         |
|-------------------------------|-------------------------------|--------|-------------------------------|--------|-----------|---------|
| Local Historic District Tract | 143                           | (73.0) | 53                            | (27.0) | 196       | (100.0) |
| Undesignated Tracts           | 664                           | (31.6) | 1440                          | (68.4) | 2104      | (100.0) |

\* Older neighborhoods are those where more than half of the buildings were older than 30 years in 1970; newer neighborhoods have more than half of the buildings constructed within 30 years of 1970.

<sup>6</sup> I checked the robustness of the five percent threshold with alternative thresholds of 1 and 10 percent. While there are 271 census tracts that include an area of local historic districts within their boundaries, the number of LHD tracts is 242 tracts at the 1% threshold, 196 tracts at the 5% threshold, and 160 tracts at the 10% threshold. Despite the different number of LHD tracts by the thresholds, the results from the robustness tests remained consistent across all thresholds.

<sup>7</sup> A property is typically considered eligible for historic designation if it is at least 50 years old, which is a sufficient time “to develop historical perspective and to evaluate significance” (National Park Service, n.d.). Despite the 50-year rule, this study uses the 30-year threshold for identifying older neighborhoods because the 1970 census data groups all structures built before 1939 into a single category.

<sup>8</sup> Both average and median ratios of older buildings (older than 30 years) in the LHD tracts in 1970 are 62%.

**Table 5**

Z-score means across clusters.

| Neighborhood Types                   | Nonfamily/Lower | Hispanic/Lower | Black/Lower | Mixed/Moderate | Nonfamily/Moderate | Professional/Upper |
|--------------------------------------|-----------------|----------------|-------------|----------------|--------------------|--------------------|
| (No. of LHD tracts, %)               | (222, 26%)      | (204, 24%)     | (83, 10%)   | (84, 10%)      | (212, 25%)         | (53, 6%)           |
| (No. of UDG tracts, %)               | (855, 21%)      | (913, 23%)     | (438, 11%)  | (673, 17%)     | (817, 21%)         | (288, 7%)          |
| (No. of total tracts, %)             | (1707, 12%)     | (2585, 19%)    | (1427, 10%) | (3532, 26%)    | (2740, 20%)        | (1,809, 13%)       |
| <i>Demographic Characteristics</i>   |                 |                |             |                |                    |                    |
| White%                               | −0.251          | −0.680         | −1.737      | 0.260          | 0.702              | 1.007              |
| Black%                               | −0.011          | −0.112         | 2.372       | −0.329         | −0.323             | −0.568             |
| Hispanic%                            | 0.106           | 1.473          | −0.050      | −0.133         | −0.667             | −0.895             |
| Immigrant%                           | 0.756           | 1.054          | −0.502      | −0.192         | −0.462             | −0.749             |
| Child%                               | −0.830          | 0.988          | 0.976       | 0.197          | −1.155             | −0.036             |
| Adult%                               | 0.541           | −0.635         | −0.857      | −0.180         | 1.183              | −0.367             |
| Aging%                               | 0.413           | −0.556         | −0.292      | −0.025         | 0.046              | 0.616              |
| Nonfamily%                           | 1.147           | −0.708         | −0.487      | −0.462         | 1.148              | −0.522             |
| <i>Socioeconomic Characteristics</i> |                 |                |             |                |                    |                    |
| High School%                         | −0.015          | 0.140          | 0.554       | 0.645          | −0.726             | −0.785             |
| College%                             | −0.332          | −1.011         | −0.888      | −0.212         | 1.020              | 1.326              |
| Professional%                        | −0.405          | −1.064         | −0.758      | −0.077         | 0.921              | 1.256              |
| Manual%                              | −0.019          | 1.280          | 0.586       | 0.108          | −0.946             | −1.049             |
| Unemployment%                        | 0.725           | 0.433          | 0.945       | −0.347         | −0.434             | −0.713             |
| Poverty%                             | 0.928           | 0.732          | 0.811       | −0.545         | −0.344             | −0.978             |
| HH Income                            | −0.839          | −0.616         | −0.583      | 0.052          | 0.169              | 1.775              |
| <i>Housing Characteristics</i>       |                 |                |             |                |                    |                    |
| Vacant%                              | 0.734           | −0.121         | 0.257       | −0.469         | 0.386              | −0.391             |
| RenterOcc%                           | 1.080           | 0.369          | −0.058      | −0.537         | 0.575              | −1.323             |
| Singleunit%                          | −1.155          | −0.088         | 0.289       | 0.558          | −0.787             | 1.090              |

\* Bolds indicate  $|Z\text{-score}| > 0.8$ ; italics indicate  $0.8 \geq |Z\text{-score}| \geq 0.5$ .

demographic and socioeconomic characteristics with median household income levels: *Nonfamily/Lower*, *Hispanic/Lower*, *Black/Lower*, *Mixed/Moderate*, *Nonfamily/Moderate*, and *Professional/Upper* (Table 5).

The three lower-income neighborhood types, *Nonfamily/Lower*, *Hispanic/Lower*, and *Black/Lower*, shared similar socioeconomic features of higher poverty and unemployment rates and a lower ratio of college-educated professionals. However, there are apparent differences in their demographic and housing characteristics. *Nonfamily/Lower* neighborhoods are distinguished by a higher proportion of nonfamily households than other neighborhood types. They also show a predominance of renter-occupied and multiunit housing with a higher vacancy rate. Although the other two lower-income clusters have a higher prevalence of families with children and manual laborers, their racial/ethnic compositions are divergent. *Hispanic/Lower* neighborhoods have the highest average z-scores for the proportion of Hispanic and immigrant populations, whereas *Black/Lower* neighborhoods have the highest for African Americans.

In the moderate-income categories, *Mixed/Moderate* neighborhoods are racially more heterogeneous than the other neighborhoods since the z-scores for any race and ethnicity are not significant. Nonfamily households are less prevalent in these neighborhoods, while they have the highest average score in the proportion of residents who completed high school but no college. These neighborhoods contain a slightly larger proportion of single-family housing with high levels of homeownership. Additionally, the most significant characteristics of *Nonfamily/Moderate* neighborhoods include a higher proportion of nonfamily adult households without children and a larger share of college-graduated professionals. They also score relatively higher in the proportion of multifamily housing and renters. As the only upper-income neighborhood type, *Professional/Upper* neighborhoods are very distinctive with exceptionally high median household incomes, accompanied by the highest shares of white populations, college graduates, and professionals. They also have the largest share of owner-occupied units and single-family housing.

#### 4.2. Neighborhood transitions

Fig. 1 illustrates temporal changes in the distributions of neighborhood types within LHD and UDG tracts between 1970 and 2020. The most prominent patterns across LHD tracts are the increase in *Nonfamily/Moderate* neighborhoods from 13% in 1970 to 38% in 2020 and the decreases in *Nonfamily/Lower* and *Black/Lower* neighborhoods from 41% to 15% in 1970 to 24% and 3% in 2020. Although these patterns are significant in UDG tracts, the extent of changes is slightly less, with a 19% increase in *Nonfamily/Moderate* neighborhoods and a 14% decrease in *Nonfamily/Lower* neighborhoods. Meanwhile, the proportions of *Hispanic/Lower* and *Mixed/Moderate* types in LHD tracts fluctuated during the study period, but there was no change in the proportion between 1970 and 2020. In UDG tracts, a modest increase in *Hispanic/Lower* neighborhoods from 16% to 23% and a slight decrease in *Mixed/Moderate* neighborhoods from 22% to 15% are observed. Lastly, the *Professional/Upper* neighborhood type has gradually grown from 4% to 8% in LHD tracts and from 3% to 12% in UDG tracts.

To compare the patterns of neighborhood change between LHD and UDG tracts, I analyzed the transitions of each census tract between neighborhood types. Table 6 shows the aggregated neighborhood transitions between census years during the study period, capturing transitions between each consecutive census year (i.e., 1970–1980, 1980–1990, etc.) for each census tract.<sup>9</sup> The most notable distinction is that LHD tracts are less stable than UDG tracts regardless of neighborhood types. In LHD tracts, *Nonfamily/Lower* and *Mixed/Moderate* had high transition rates of over 30 percent, compared to UDG tracts, which were in the mid 20s. Most of the transitions from *Nonfamily/Lower* neighborhoods in LHD tracts include upward movement to *Nonfamily/*

<sup>9</sup> For instance, a neighborhood classified as *Nonfamily/Lower* in 1970 may transition to *Hispanic/Lower* in 1980 and then to *Black/Lower* in 1990. In this case, a transition from *Nonfamily/Lower* to *Hispanic/Lower* and another transition from *Hispanic/Lower* to *Black/Lower* are separately counted and aggregated.

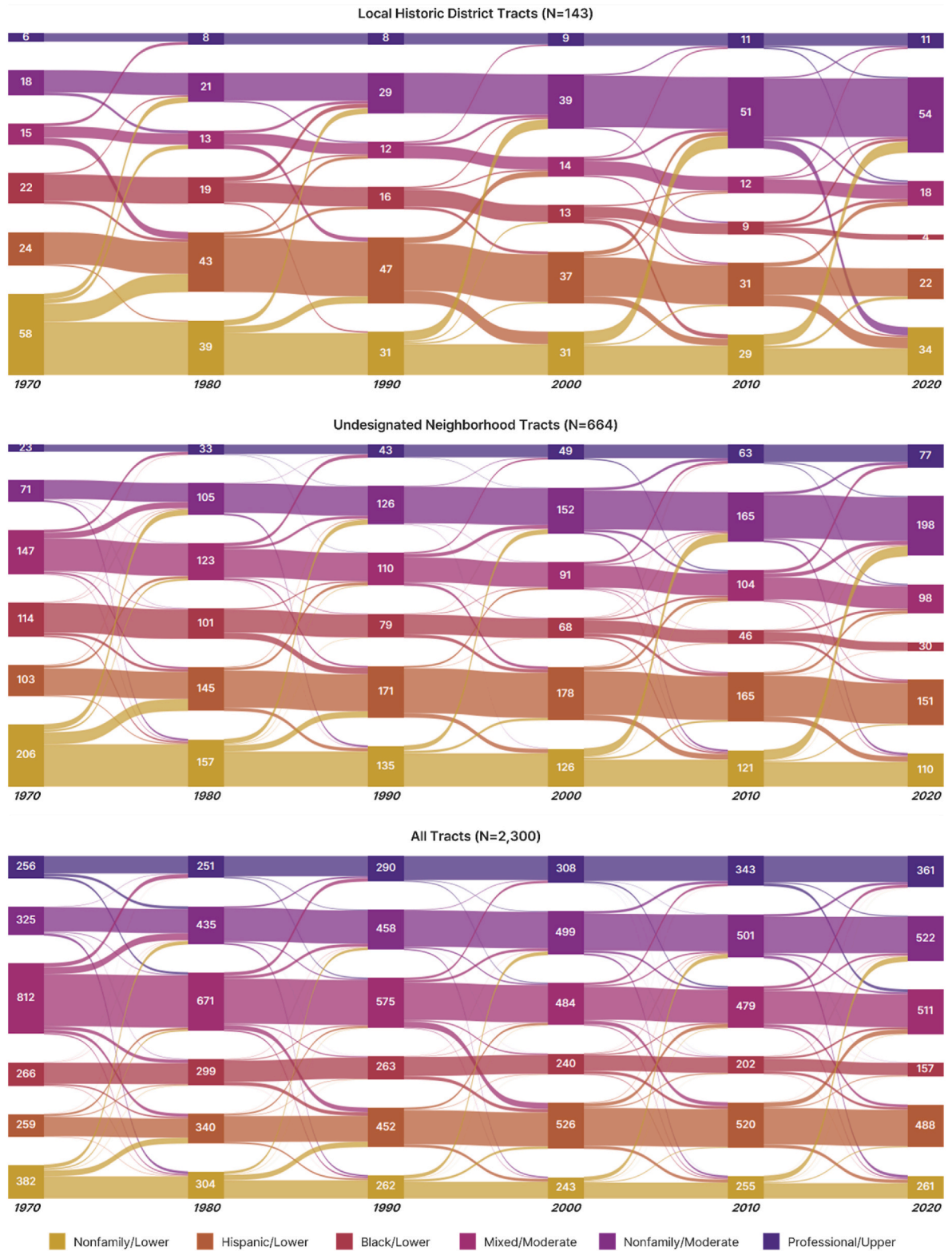


Fig. 1. The number of neighborhood types and their transitions across decades.

*Moderate* neighborhoods and horizontal movement to *Hispanic/Lower* neighborhoods as a similar income-level cluster. Meanwhile, the transition patterns from *Mixed/Moderate* neighborhoods diverge between upgrading, downgrading, and transitioning to the similar income-level cluster. Additionally, the major patterns from the other two lower-income neighborhoods in LHD tracts are transitions to similar income level clusters, particularly significant from *Hispanic/Lower* to

*Nonfamily/Lower*, and neighborhood ascent to moderate-income types. *Professional/Upper* and *Nonfamily/Moderate* neighborhoods had lower transition rates than the other clusters. All *Professional/Upper* neighborhoods in LHD tracts remain in their cluster except for two downgrading cases to moderate-income neighborhoods. Also, there was no direct transition between the upper- and lower-income clusters in LHD tracts. With the high retention rate of the *Nonfamily/Moderate* type

**Table 6**  
Aggregated Neighborhood Transitions across decades.

| Neighborhood Type<br>from 1970 to 2020 | Transition<br>Rates | Neighborhood type in subsequent Census year (%) |                    |                 |                    |                        |                        |
|----------------------------------------|---------------------|-------------------------------------------------|--------------------|-----------------|--------------------|------------------------|------------------------|
|                                        |                     | Nonfamily/<br>Lower                             | Hispanic/<br>Lower | Black/<br>Lower | Mixed/<br>Moderate | Nonfamily/<br>Moderate | Professional/<br>Upper |
| <i>LHD tracts</i>                      |                     |                                                 |                    |                 |                    |                        |                        |
| Nonfamily/Lower                        | 30.9%               | 130                                             | 22                 | 1               | 3                  | 32                     | 0                      |
| Hispanic/Lower                         | 20.9%               | 23                                              | 144                | 2               | 10                 | 3                      | 0                      |
| Black/Lower                            | 27.8%               | 5                                               | 5                  | 57              | 5                  | 7                      | 0                      |
| Mixed/Moderate                         | 31.8%               | 0                                               | 9                  | 0               | 45                 | 7                      | 5                      |
| Nonfamily/<br>Moderate                 | 8.9%                | 6                                               | 0                  | 1               | 5                  | 144                    | 2                      |
| Professional/Upper                     | 4.8%                | 0                                               | 0                  | 0               | 1                  | 1                      | 40                     |
| <i>UDG tracts</i>                      |                     |                                                 |                    |                 |                    |                        |                        |
| Nonfamily/Lower                        | 27.4%               | 541                                             | 79                 | 1               | 11                 | 113                    | 0                      |
| Hispanic/Lower                         | 15.6%               | 64                                              | 643                | 4               | 43                 | 8                      | 0                      |
| Black/Lower                            | 23.8%               | 10                                              | 61                 | 311             | 19                 | 5                      | 2                      |
| Mixed/Moderate                         | 25.0%               | 8                                               | 27                 | 7               | 431                | 58                     | 44                     |
| Nonfamily/<br>Moderate                 | 10.0%               | 26                                              | 0                  | 1               | 14                 | 557                    | 21                     |
| Professional/Upper                     | 6.2%                | 0                                               | 0                  | 0               | 8                  | 5                      | 198                    |
| <i>All tracts</i>                      |                     |                                                 |                    |                 |                    |                        |                        |
| Nonfamily/Lower                        | 30.1%               | 1011                                            | 175                | 8               | 32                 | 219                    | 1                      |
| Hispanic/Lower                         | 15.9%               | 125                                             | 1763               | 42              | 155                | 12                     | 0                      |
| Black/Lower                            | 19.3%               | 17                                              | 158                | 1025            | 50                 | 17                     | 3                      |
| Mixed/Moderate                         | 23.5%               | 59                                              | 230                | 74              | 2310               | 180                    | 168                    |
| Nonfamily/<br>Moderate                 | 13.0%               | 112                                             | 0                  | 11              | 78                 | 1930                   | 87                     |
| Professional/Upper                     | 10.6%               | 1                                               | 0                  | 1               | 95                 | 57                     | 1294                   |

**Table 7**  
The first ten sequences of neighborhood transitions.

| Neighborhood Type from 1970 to 2020 |   |                    | Count | Ratio |
|-------------------------------------|---|--------------------|-------|-------|
| <i>LHD tracts</i>                   |   |                    |       |       |
| Nonfamily/Lower                     | → | Nonfamily/Moderate | 26    | 18.2% |
| Nonfamily/Lower                     | → | Nonfamily/Lower    | 25    | 17.5% |
| Nonfamily/Moderate                  | → | Nonfamily/Moderate | 14    | 9.8%  |
| Hispanic/Lower                      | → | Hispanic/Lower     | 11    | 7.7%  |
| Mixed/Moderate                      | → | Mixed/Moderate     | 11    | 7.7%  |
| Nonfamily/Lower                     | → | Hispanic/Lower     | 7     | 4.9%  |
| Black/Lower                         | → | Nonfamily/Moderate | 7     | 4.9%  |
| Hispanic/Lower                      | → | Nonfamily/Lower    | 6     | 4.2%  |
| Hispanic/Lower                      | → | Nonfamily/Moderate | 6     | 4.2%  |
| Professional/Upper                  | → | Professional/Upper | 6     | 4.2%  |
| <i>UDG tracts</i>                   |   |                    |       |       |
| Nonfamily/Lower                     | → | Nonfamily/Moderate | 93    | 14.0% |
| Nonfamily/Lower                     | → | Nonfamily/Lower    | 87    | 13.1% |
| Mixed/Moderate                      | → | Mixed/Moderate     | 65    | 9.8%  |
| Hispanic/Lower                      | → | Hispanic/Lower     | 58    | 8.7%  |
| Black/Lower                         | → | Hispanic/Lower     | 56    | 8.4%  |
| Nonfamily/Moderate                  | → | Nonfamily/Moderate | 49    | 7.4%  |
| Mixed/Moderate                      | → | Professional/Upper | 44    | 6.6%  |
| Nonfamily/Lower                     | → | Hispanic/Lower     | 26    | 3.9%  |
| Black/Lower                         | → | Black/Lower        | 26    | 3.9%  |
| Mixed/Moderate                      | → | Nonfamily/Moderate | 24    | 3.6%  |
| <i>All tracts</i>                   |   |                    |       |       |
| Mixed/Moderate                      | → | Mixed/Moderate     | 381   | 16.6% |
| Nonfamily/Moderate                  | → | Nonfamily/Moderate | 220   | 9.6%  |
| Professional/Upper                  | → | Professional/Upper | 192   | 8.3%  |
| Nonfamily/Lower                     | → | Nonfamily/Moderate | 164   | 7.1%  |
| Mixed/Moderate                      | → | Hispanic/Lower     | 160   | 7.0%  |
| Nonfamily/Lower                     | → | Nonfamily/Lower    | 155   | 6.7%  |
| Hispanic/Lower                      | → | Hispanic/Lower     | 153   | 6.7%  |
| Mixed/Moderate                      | → | Professional/Upper | 127   | 5.5%  |
| Black/Lower                         | → | Hispanic/Lower     | 115   | 5.0%  |
| Black/Lower                         | → | Black/Lower        | 90    | 3.9%  |

in LHD tracts, another significant pattern is a large number of transitions from other clusters. The most frequent transitions to *Nonfamily/Moderate* are from *Nonfamily/Lower* and *Black/Lower*, signifying upward

movements, followed by horizontal transitions from *Mixed/Moderate*.

**Table 7** displays the most frequent trajectories of neighborhood transitions from 1970 to 2020 along with their most common longitudinal sequences in each subsequent census year (for more details, please see [Appendix B](#)). Consistent with previous studies ([Delmelle, 2015](#); [Foote & Walter, 2017](#); [Wei & Knox, 2014](#)), the most commonly observed trajectories are neighborhoods remaining in the same type at both the beginning and end of the 50-year study duration, representing 50% of LHD tracts. However, less than half of these stability trajectories in *Nonfamily/Lower* and *Mixed/Moderate* have no changes during the entire period, suggesting the instability of these neighborhood types. In addition, neighborhood upgrades from the lower-income neighborhood types to *Nonfamily/Moderate* neighborhoods comprise 27% of LHD tracts, compared to 19% of UDG tracts. Their longitudinal sequences in LHD tracts indicate that upgrades from *Nonfamily/Lower* and *Black/Lower* neighborhoods occurred continuously over the entire 50-year period, while ascents from *Hispanic/Lower* neighborhoods were mostly concentrated in the 2000s. In contrast, there are only two downgrading cases to the lower-income neighborhoods. In the last pattern of transitions between the same income-level clusters, the sequences between *Nonfamily/Lower* and *Hispanic/Lower* neighborhoods were significant, particularly from *Nonfamily/Lower* to *Hispanic/Lower* in the 1970s and from *Hispanic/Lower* to *Nonfamily/Lower* in the 2010s.

#### 4.3. Neighborhood change in local historic districts

Based on the analysis of neighborhood transitions across the typologies, I further investigate distinctions in their patterns by phases of district designation. **Table 8** displays the patterns of the neighborhood transitions in LHD tracts categorized into three timeframes: before (pre-), during, and after (post-) designation. Additionally, each transition from one neighborhood type to another between census years is classified into upgrading, downgrading, horizontal transitions, and stable cases. While transitions between neighborhood types with different household income z-scores indicate upgrading or downgrading, changes between similar income clusters are assigned to horizontal transitions. Stability cases refer to no change in neighborhood types. While LHD



**Table 8**

Aggregated neighborhood transition patterns in LHD tracts.

|                    | Upgrading |         | Downgrading |        | Horizontal |         | Stable |         |
|--------------------|-----------|---------|-------------|--------|------------|---------|--------|---------|
| Pre-designation    | 23        | (7.0%)  | 9           | (2.7%) | 33         | (10.1%) | 263    | (80.2%) |
| During designation | 16        | (11.2%) | 3           | (1.4%) | 13         | (9.1%)  | 112    | (78.3%) |
| Post-designation   | 28        | (11.5%) | 7           | (2.9%) | 24         | (9.8%)  | 185    | (75.8%) |

tracts were more likely to remain in the same cluster before designation than during and after designation, the proportions of upward movement in LHD neighborhoods were higher during and after designation. Across the neighborhood types, these patterns are more prevalent in lower-income neighborhoods with a higher probability of upward transitions to moderate-income types during and after designation phases. Conversely, Nonfamily/Moderate and Professional/Upper neighborhood types mostly remained stable (for more details, please see Appendix C).

## 5. Discussion

This study explores the trajectories of neighborhood transitions in older neighborhoods within the context of growth, focusing on how the designation of local historic districts intersects with neighborhood socioeconomic change. A partitioning cluster analysis of longitudinal census data across the selected eight cities in the Western region identified six specific neighborhood types, each with distinct demographic, socioeconomic, and housing characteristics: *Nonfamily/Lower, Hispanic/Lower, Black/Lower, Mixed/Moderate, Nonfamily/Moderate, and Professional/Upper*. This classification enabled an in-depth examination of temporal transitions and longitudinal sequences in local historic districts to answer three research questions: (1) What are the transition patterns and neighborhood characteristics of local historic districts compared to undesignated neighborhoods? (2) How does the association of locally designated historic districts with neighborhood stability vary across different neighborhood types? (3) How do the patterns of neighborhood change differ by the phases of district designation: before, during, and after designation?

### 5.1. Catalytic role of local historic districts for neighborhood ascent

The analysis of neighborhood transition patterns in the context of growth reveals that local historic districts have a higher likelihood of upward transitions than undesignated neighborhoods. Both local historic districts and undesignated neighborhoods display a decrease in lower-income neighborhoods and an increase in moderate- and upper-income neighborhoods, with upward movements appearing more frequently in local historic districts (Fig. 1). The greater prevalence of the upgrading patterns suggests the potential catalytic role of local historic districts for neighborhood change, consistent with the findings of previous research that the creation of local historic districts leads to changes in socioeconomic status (McCabe & Ellen, 2016) and race (Kinahan & Ruther, 2020). However, the proportion of lower-income neighborhoods is initially higher than the other neighborhood types, which may influence the differences in the prevalence of upgrading patterns based on the income level of neighborhood types.

The most significant pattern of neighborhood change in local historic districts is the substantial increase in *Nonfamily/Moderate* neighborhoods, contributing to the widespread upward movements. The common longitudinal sequences to the *Nonfamily/Moderate* type are neighborhood ascents from the three lower-income neighborhood types, and transitions from *Nonfamily/Lower* neighborhoods are the largest (Table 7). This result suggests that the creation of local historic districts was likely to serve as a revitalization strategy within the context of growth (Hurley, 2010; Kinahan & Mawhorter, 2021). As local historic

districts are often designated in downtown and inner-city neighborhoods where older buildings are concentrated, their designation can facilitate preservation activities such as the rehabilitation and adaptive reuse of historic buildings (Birch, 2002, 2005; Ryberg-Webster, 2013). As a result, this may contribute to attracting specific population groups like creative workers or young professionals seeking authentic character, walkability, and human-scale architecture in historic neighborhoods (Allison & Peters, 2011; Florida, 2002).

### 5.2. Disparities in neighborhood stability across local historic districts

The results from the transition rate analysis indicate that local historic districts were generally less stable than undesignated neighborhoods within the context of growth (Table 6), which challenges the underlying expectation of preservation policies to promote neighborhood stability (Birch & Roby, 1984; Rose, 1981). In particular, all three lower-income neighborhood types in local historic districts display higher transition rates than those in undesignated neighborhoods, while moderate- and upper-income neighborhoods generally had similar transition rates. These results imply that the impacts of local historic districts on neighborhood socioeconomic change could be more intensified in lower-income neighborhoods, thereby potentially facing a higher risk of gentrification and displacement attributed to historic district designation (Oba & Noonan, 2020; Zahirovic-Herbert & Chatterjee, 2012). These findings suggest the need to consider the potential impacts of local historic districts on neighborhood stability and encourage additional policies and programs, such as creating affordable housing (Cohen, 1998; Passell, 2021; Ryberg-Webster, 2019) and protecting intangible cultural heritage (Buckley & Graves, 2016; Kaufman, 2013), to mitigate the negative impacts.

Moreover, a much lower transition rate in local historic districts categorized as *Nonfamily/Moderate* and *Professional/Upper* neighborhood types compared to lower-income neighborhoods suggests that the designation of local historic districts may contribute to exacerbating socio-spatial inequalities in urban areas, given the recent trend toward more historic district designations where upper-income neighborhoods are concentrated (Choi et al., 2025). This difference indicates that higher-income neighborhoods can benefit from increased stability ensured by honorary status and regulatory controls of local historic districts (Heintzelman & Altieri, 2013), whereas lower-income neighborhoods remain more vulnerable to socioeconomic transitions. This disparate influence of historic district designation on neighborhood stability could reduce affordability and social diversity (Talen et al., 2015) and aggravate the extent of segregation and exclusiveness (Bologna Pavlik & Zhou, 2023). Therefore, it is possible that the designation of local historic districts may serve as exclusionary zoning since regulatory controls can restrict changes in the neighborhoods, thereby excluding the influx of new households (Fein, 1985).

### 5.3. Different transition patterns by the phases of district designation

Exploring the trajectories of neighborhood change only in local historic districts indicates obvious distinctions in the transition patterns between pre-designation, during designation, and post-designation (Table 8). These different patterns of neighborhood change across the phases of district designation imply a potential connection of the designation process with endogenous factors, such as neighborhood characteristics, local contexts, and collective actions (Ahlfeldt et al., 2017; Been et al., 2016; Noonan & Krupka, 2011; Zhou, 2021). In particular, neighborhoods with local historic districts were more likely to remain in the same neighborhood type during the pre-designation period and then experience upgrading movements during and after designation, especially frequently observed in lower-income neighborhoods. These longitudinal trends within local historic districts align with the traditional rent gap theory (Smith, 1979), which postulates that the decline of inner-city neighborhoods is a prerequisite for gentrification

because the depreciation and devaluation of capital create favorable conditions for profitable reinvestment via rehabilitation and redevelopment. Although some may regard many of the stability cases in lower-income neighborhoods as a sign of neighborhood stabilization, others may view this result as part of the process of increasing the rent gap to maximize profits via investment in older neighborhoods.

Moreover, a larger probability of neighborhood upward movements during or after designation may reflect that the designation of historic districts is more likely to attract those with greater wealth. The designation status can be regarded as the certification of a better-maintained property that requires less investment in its rehabilitation and restoration since historic designation typically requires high standards of architectural integrity as well as historical significance (Michael, 2014). Conversely, old buildings in under-resourced neighborhoods that may have suffered from gradual deterioration and neglect cannot satisfy the integrity requirements (Ryberg-Webster, 2017). However, the trends observed within local historic districts could merely reflect general patterns of neighborhood change in the studied region, raising questions about the extent to which district designation influences these observed trends. Since the 1990s, the higher likelihood of upward movement from lower-income neighborhoods within local historic districts during and after designation has paralleled a regional trend of socioeconomic ascent, making it challenging to distinguish the direct effects of historic district designation from broader regional transition patterns.

#### 5.4. Study limitations and future research directions

The results from longitudinal analysis shed light on six census years of neighborhood transitions in local historic districts in comparison to undesignated neighborhoods within the growing context, but the research protocol used in this study has some limitations. First, multivariate cluster approach used in this study is a descriptive analysis that is not able to examine the causal relationship between the designation of local historic districts and neighborhood change (Delmelle, 2015). There is a need for additional research that employs inferential statistical methods to investigate how the creation of local historic districts affects observed neighborhood transition patterns. Second, while this study focuses on the trends of neighborhood change in the context of growth by narrowing down the study area to cities in the Western region, further studies could broaden a larger sample of cities across different regions. Exploring cities within distinct contexts is more likely to display diverse trajectories of neighborhood change (Delmelle, 2015; Wei & Knox, 2014), and furthermore, it allows highlighting the varying roles and impacts of local historic districts according to growing trends of cities and metropolitan areas. Third, although this study examines general patterns across multiple cities with very similar policy frameworks of local historic districts, there is a possibility that variations in preservation ordinances about local historic districts between cities might lead to different trajectories of neighborhood change. Future research could compare several individual cities with different designation procedures and regulations for local historic districts to examine how different policy frameworks are related to the patterns of neighborhood change. Lastly, I identified clusters of neighborhoods within the city limits because local historic districts are legally based on municipal preservation ordinances, as opposed to the larger scope of Metropolitan

Statistical Areas typically used in previous neighborhood typology research. This approach could underrepresent a specific neighborhood type with suburban characteristics and not fully capture the broader trends of neighborhood transitions at the metropolitan level.

## 6. Conclusion

This study explored the complex dynamics between local historic districts and neighborhood socioeconomic change within the context of growth by utilizing a k-means cluster approach with longitudinal data on multiple neighborhood characteristics. This analysis reveals the distinct trajectories of neighborhood transitions over 50 years in local historic districts, highlighting variations across different neighborhood types and phases of district designation. The study found that local historic districts tended to be more unstable than undesignated neighborhoods, with more prevalent upward transitions in lower-income neighborhoods within local historic districts, especially during and after designation. These results from multivariate longitudinal analysis address a gap in the literature by providing empirical evidence on the potential catalytic function of historic district designation on neighborhood change.

Moreover, this study contributes to the recent growing discourse on equitable and inclusive historic preservation, emphasizing social justice, cultural diversity, and sustainable development in preservation practices and policies (Arlotta & Avrami, 2020; Cheong, 2021; Minner, 2016; Wells & Stiefel, 2019). Understanding the potential gentrification and displacement risks associated with historic district designation can encourage preservation planners and landmark committees to consider the equitable distribution of costs and outcomes from preservation policies. It highlights the need for a more integrated approach that incorporates housing policies and planning tools in the designation process to mitigate the socioeconomic impacts of district designation. In addition, the large variability in neighborhood trajectories across different neighborhood types suggests the need for anticipatory research on the socioeconomic transformation in older neighborhoods that could potentially be designated as local historic districts. Analyzing the neighborhood characteristics within metropolitan and regional contexts can help predict the potential impacts of local historic districts on neighborhood change and identify strategies against the negative effects in advance.

## Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgments

This paper is based on my dissertation at the University of Washington, and I would like to thank my committee, Manish Chalana, Jeffrey Karl Ochsner, Rebecca J. Walter, and Dan Abramson, for their feedback. Also, I appreciate the anonymous reviewers for the insightful comments on the paper.

## Appendix A. Summary of five and seven cluster options

**Table A.1**

Z-score means across five clusters

|                          | Type 1      | Type 2      | Type 3      | Type 4      | Type 5      |
|--------------------------|-------------|-------------|-------------|-------------|-------------|
| (No. of LHD tracts; %)   | (83, 10%)   | (62, 7%)    | (287, 33%)  | (93, 11%)   | (333, 19%)  |
| (No. of UDG tracts; %)   | (680, 17%)  | (365, 9%)   | (1165, 29%) | (478, 12%)  | (1296, 33%) |
| (No. of total tracts; %) | (2959, 21%) | (1520, 11%) | (3664, 27%) | (3565, 26%) | (2092, 15%) |
| <i>Demographic</i>       |             |             |             |             |             |
| White%                   | −0.653      | −1.707      | 0.463       | 0.206       | 1.003       |
| Black%                   | −0.118      | 2.301       | −0.237      | −0.315      | −0.554      |
| Hispanic%                | 1.305       | −0.022      | −0.488      | −0.053      | −0.884      |
| Immigrant%               | 1.152       | −0.447      | −0.154      | −0.160      | −0.762      |
| Child%                   | 0.729       | 0.929       | −1.159      | 0.262       | −0.122      |
| Adult%                   | −0.474      | −0.812      | 1.079       | −0.222      | −0.250      |
| Aging%                   | −0.412      | −0.277      | 0.175       | −0.059      | 0.577       |
| Nonfamily%               | −0.399      | −0.440      | 1.242       | −0.521      | −0.404      |
| <i>Socioeconomic</i>     |             |             |             |             |             |
| High School%             | 0.055       | 0.584       | −0.511      | 0.682       | −0.770      |
| College%                 | −0.929      | −0.906      | 0.652       | −0.278      | 1.304       |
| Professional%            | −1.001      | −0.783      | 0.569       | −0.146      | 1.238       |
| Manual%                  | 1.111       | 0.615       | −0.744      | 0.191       | −1.040      |
| Unemployment%            | 0.506       | 1.052       | −0.133      | −0.321      | −0.701      |
| Poverty%                 | 0.851       | 0.864       | 0.016       | −0.531      | −0.956      |
| HH Income                | −0.696      | −0.610      | −0.161      | 0.032       | 1.654       |
| <i>Housing</i>           |             |             |             |             |             |
| Vacant%                  | 0.040       | 0.304       | 0.501       | −0.481      | −0.336      |
| RenterOcc%               | 0.606       | −0.027      | 0.762       | −0.571      | −1.198      |
| Singleunit%              | −0.366      | 0.251       | −0.948      | 0.609       | 0.959       |

\* Bolds indicate  $|Z\text{-score}| > 0.8$ ; italics indicate  $0.8 \geq |Z\text{-score}| \geq 0.5$ .

**Table A.2**

Z-score means across seven clusters

|                          | Type 1     | Type 2      | Type 3      | Type 4      | Type 5      | Type 6      | Type 7      |
|--------------------------|------------|-------------|-------------|-------------|-------------|-------------|-------------|
| (No. of LHD tracts; %)   | (120, 14%) | (162, 19%)  | (87, 10%)   | (162, 19%)  | (66, 8%)    | (208, 24%)  | (53, 6%)    |
| (No. of UDG tracts; %)   | (429, 11%) | (771, 19%)  | (433, 11%)  | (692, 17%)  | (562, 14%)  | (820, 21%)  | (277, 7%)   |
| (No. of total tracts; %) | (666, 5%)  | (2281, 17%) | (1362, 10%) | (1936, 14%) | (3080, 22%) | (2759, 20%) | (1716, 12%) |
| <i>Demographic</i>       |            |             |             |             |             |             |             |
| White%                   | −0.111     | −0.724      | −1.747      | −0.285      | 0.316       | 0.703       | 1.016       |
| Black%                   | 0.114      | −0.060      | 2.422       | −0.130      | −0.356      | −0.329      | −0.572      |
| Hispanic%                | −0.044     | 1.551       | −0.089      | 0.325       | −0.201      | −0.669      | −0.907      |
| Immigrant%               | 0.375      | 0.995       | −0.555      | 0.820       | −0.276      | −0.438      | −0.753      |
| Child%                   | −1.477     | 1.109       | 0.950       | −0.107      | 0.208       | −1.161      | −0.040      |
| Adult%                   | 0.894      | −0.740      | −0.858      | 0.184       | −0.211      | 1.158       | −0.373      |
| Aging%                   | 0.796      | −0.576      | −0.267      | −0.096      | 0.003       | 0.088       | 0.631       |
| Nonfamily%               | 1.855      | −0.840      | −0.447      | 0.345       | −0.539      | 1.147       | −0.513      |
| <i>Socioeconomic</i>     |            |             |             |             |             |             |             |
| High School%             | −0.064     | 0.204       | 0.528       | 0.104       | 0.669       | −0.715      | −0.835      |
| College%                 | −0.405     | −1.083      | −0.872      | −0.356      | −0.176      | 1.015       | 1.374       |
| Professional%            | −0.456     | −1.117      | −0.736      | −0.406      | −0.032      | 0.911       | 1.296       |
| Manual%                  | −0.113     | 1.404       | 0.550       | 0.185       | 0.079       | −0.952      | −1.078      |
| Unemployment%            | 1.665      | 0.491       | 0.959       | 0.049       | −0.389      | −0.437      | −0.715      |
| Poverty%                 | 1.626      | 0.725       | 0.828       | 0.368       | −0.621      | −0.357      | −0.981      |
| HH Income                | −1.055     | −0.604      | −0.589      | −0.579      | 0.132       | 0.164       | 1.832       |
| <i>Housing</i>           |            |             |             |             |             |             |             |
| Vacant%                  | 1.540      | −0.162      | 0.287       | 0.098       | −0.531      | 0.376       | −0.372      |
| RenterOcc%               | 1.046      | 0.229       | −0.032      | 0.863       | −0.706      | 0.579       | −1.322      |
| Singleunit%              | −1.387     | 0.075       | 0.262       | −0.774      | 0.721       | −0.797      | 1.090       |

\* Bolds indicate  $|Z\text{-score}| > 0.8$ ; italics indicate  $0.8 \geq |Z\text{-score}| \geq 0.5$ .

In order to determine the optimal number of clusters, I conducted separate k-means approaches with five, six, and seven clusters. The six-cluster option offered distinct and interpretable neighborhood typologies and a balanced distribution of samples in each cluster. In contrast, the five-cluster option resulted in broader groupings that particularly lacked sufficient differentiation in Type 3 (Table A1). Type 3 includes mixed neighborhood characteristics from *Nonfamily/Lower* and *Nonfamily/Moderate* identified in the six-cluster option. In addition, the seven-cluster option overly

separated clusters, resulting in a very small sample size for Type 1 (Table A2). *Nonfamily/Lower* neighborhood type in the six-cluster option is divided into Types 1 and 4, with Type 1 comprising only five percent of the total sample.

## Appendix B. Sequences of neighborhood transitions

**Table B.1**

The first ten sequences of neighborhood transitions

| Neighborhood Type<br>from 1970 to 2020 |   |      | Count | Ratio | Top Three Sequences (count in parentheses) |             |             |
|----------------------------------------|---|------|-------|-------|--------------------------------------------|-------------|-------------|
| LHD tracts                             |   |      |       |       |                                            |             |             |
| 1-NL                                   | → | 5-NM | 26    | 18.2% | 111115(5)                                  | 111555 (5)  | 115555 (4)  |
| 1-NL                                   | → | 1-NL | 25    | 17.5% | 111111 (11)                                | 111151 (3)  | 2 cases (2) |
| 5-NM                                   | → | 5-NM | 14    | 9.8%  | 555555 (13)                                | 544455 (1)  | –           |
| 2-HL                                   | → | 2-HL | 11    | 7.7%  | 222222 (11)                                | –           | –           |
| 4-MM                                   | → | 4-MM | 11    | 7.7%  | 444444 (2)                                 | 442444 (2)  | 7 cases (1) |
| 1-NL                                   | → | 2-HL | 7     | 4.9%  | 122222 (4)                                 | 3 cases (1) | -           |
| 3-BL                                   | → | 5-NM | 7     | 4.9%  | 335555 (2)                                 | 5 cases (1) | -           |
| 2-HL                                   | → | 1-NL | 6     | 4.2%  | 222221 (4)                                 | 212111 (1)  | 222211 (1)  |
| 2-HL                                   | → | 5-NM | 6     | 4.2%  | 222255 (3)                                 | 222155 (2)  | 222215 (1)  |
| 6-PU                                   | → | 6-PU | 6     | 4.2%  | 666666 (6)                                 | –           | –           |
| UDG tracts                             |   |      |       |       |                                            |             |             |
| 1-NL                                   | → | 5-NM | 93    | 14.0% | 111115 (21)                                | 155555 (19) | 111155 (15) |
| 1-NL                                   | → | 1-NL | 87    | 13.1% | 111111 (47)                                | 122221 (7)  | 3 cases (5) |
| 4-MM                                   | → | 4-MM | 65    | 9.8%  | 444444 (47)                                | 3 cases (1) | -           |
| 2-HL                                   | → | 2-HL | 58    | 8.7%  | 222222 (48)                                | 224222 (3)  | 7 cases (1) |
| 3-BL                                   | → | 2-HL | 56    | 8.4%  | 332222 (21)                                | 333222 (10) | 3 cases (8) |
| 5-NM                                   | → | 5-NM | 49    | 7.4%  | 555555 (43)                                | 6 cases (1) | -           |
| 4-MM                                   | → | 6-PU | 44    | 6.6%  | 446666 (9)                                 | 466666 (7)  | 2 cases (6) |
| 1-NL                                   | → | 2-HL | 26    | 3.9%  | 112222 (9)                                 | 122222 (9)  | 2 cases (2) |
| 3-BL                                   | → | 3-BL | 26    | 3.9%  | 333333 (26)                                | –           | –           |
| 4-MM                                   | → | 5-NM | 24    | 3.6%  | 455555 (9)                                 | 444445 (4)  | 2 cases (3) |
| All tracts                             |   |      |       |       |                                            |             |             |
| 4-MM                                   | → | 4-MM | 381   | 16.6% | 444444 (251)                               | 444424 (11) | 2 cases (9) |
| 5-NM                                   | → | 5-NM | 220   | 9.6%  | 555555 (201)                               | 555515 (3)  | 555565 (3)  |
| 6-PU                                   | → | 6-PU | 192   | 8.3%  | 666666 (173)                               | 656666 (5)  | 2 cases (2) |
| 1-NL                                   | → | 5-NM | 164   | 7.1%  | 111115 (38)                                | 155555 (36) | 111555 (21) |
| 4-MM                                   | → | 2-HL | 160   | 7.0%  | 442222 (39)                                | 442222 (25) | 422222 (24) |
| 1-NL                                   | → | 1-NL | 155   | 6.7%  | 111111 (73)                                | 122221 (12) | 111151 (9)  |
| 2-HL                                   | → | 2-HL | 153   | 6.7%  | 222222 (127)                               | 222242 (3)  | 224222 (3)  |
| 4-MM                                   | → | 6-PU | 127   | 5.5%  | 466666 (28)                                | 446666 (22) | 444446 (17) |
| 3-BL                                   | → | 2-HL | 115   | 5.0%  | 332222 (34)                                | 333222 (23) | 333332 (21) |
| 3-BL                                   | → | 3-BL | 90    | 3.9%  | 333333 (83)                                | 322233 (2)  | 333233 (2)  |

\* Each neighborhood type is labeled as 1-NL (Nonfamily/Lower), 2-HL (Hispanic/Lower), 3-BL (Black/Lower), 4-MM (Mixed/Moderate), 5-NM (Nonfamily/Moderate), and 6-PU (Professional/Upper).

## Appendix C. Transitions by the phases of district designation

**Table C.1**

Aggregated neighborhood transitions in LHD tracts

| Neighborhood Type<br>from 1970 to 2020 | Transition<br>Rates | Neighborhood type in subsequent Census year (%) |                    |                 |                    |                        |                        |
|----------------------------------------|---------------------|-------------------------------------------------|--------------------|-----------------|--------------------|------------------------|------------------------|
|                                        |                     | Nonfamily/<br>Lower                             | Hispanic/<br>Lower | Black/<br>Lower | Mixed/<br>Moderate | Nonfamily/<br>Moderate | Professional/<br>Upper |
| <i>Pre-designation</i>                 |                     |                                                 |                    |                 |                    |                        |                        |
| Nonfamily/Lower                        | 40.3%               | 40                                              | 14                 | 1               | 3                  | 9                      | 0                      |
| Hispanic/Lower                         | 12.9%               | 5                                               | 81                 | 2               | 5                  | 0                      | 0                      |
| Black/Lower                            | 15.6%               | 1                                               | 4                  | 38              | 1                  | 1                      | 0                      |
| Mixed/Moderate                         | 34.0%               | 0                                               | 9                  | 0               | 31                 | 3                      | 4                      |
| Nonfamily/<br>Moderate                 | 5.5%                | 0                                               | 0                  | 0               | 3                  | 52                     | 0                      |
| Professional/Upper                     | 0.0%                | 0                                               | 0                  | 0               | 0                  | 0                      | 21                     |
| <i>During designation</i>              |                     |                                                 |                    |                 |                    |                        |                        |
| Nonfamily/Lower                        | 24.3%               | 28                                              | 5                  | 0               | 0                  | 4                      | 0                      |
| Hispanic/Lower                         | 28.2%               | 5                                               | 28                 | 0               | 4                  | 2                      | 0                      |
| Black/Lower                            | 27.8%               | 0                                               | 0                  | 13              | 2                  | 3                      | 0                      |
| Mixed/Moderate                         | 36.4%               | 0                                               | 0                  | 0               | 7                  | 3                      | 1                      |
| Nonfamily/<br>Moderate                 | 3.6%                | 1                                               | 0                  | 0               | 0                  | 27                     | 0                      |

(continued on next page)



Table C.1 (continued)

| Neighborhood Type<br>from 1970 to 2020 | Transition<br>Rates | Neighborhood type in subsequent Census year (%) |                    |                 |                    |                        |                        |
|----------------------------------------|---------------------|-------------------------------------------------|--------------------|-----------------|--------------------|------------------------|------------------------|
|                                        |                     | Nonfamily/<br>Lower                             | Hispanic/<br>Lower | Black/<br>Lower | Mixed/<br>Moderate | Nonfamily/<br>Moderate | Professional/<br>Upper |
| Professional/Upper                     | 10.0%               | 0                                               | 0                  | 0               | 0                  | 1                      | 9                      |
| <i>Post-designation</i>                |                     |                                                 |                    |                 |                    |                        |                        |
| Nonfamily/Lower                        | 26.2%               | 62                                              | 3                  | 0               | 0                  | 19                     | 0                      |
| Hispanic/Lower                         | 30.0%               | 13                                              | 35                 | 0               | 1                  | 1                      | 0                      |
| Black/Lower                            | 62.5%               | 4                                               | 1                  | 6               | 2                  | 3                      | 0                      |
| Mixed/Moderate                         | 12.5%               | 0                                               | 0                  | 0               | 7                  | 1                      | 0                      |
| Nonfamily/<br>Moderate                 | 13.3%               | 5                                               | 0                  | 1               | 2                  | 65                     | 2                      |
| Professional/Upper                     | 9.1%                | 0                                               | 0                  | 0               | 1                  | 0                      | 10                     |

Table C1 displays the aggregated neighborhood transitions in LHD tracts using the three timeframe categories. Particularly, the distribution of neighborhood types during designation reflects the neighborhood characteristics of local historic districts. Two-thirds of neighborhoods in this period were lower-income neighborhood types, and many of them were *Hispanic/Lower* (29% of the total) and *Nonfamily/Lower* (24% of the total). These lower-income neighborhoods were less stable than other neighborhoods through upward transitions to moderated-income neighborhoods or horizontal transitions between them. Additionally, *Nonfamily/Moderate* was the next largest neighborhood cluster during designation (20% of the total), and most of them were stable.

*Nonfamily/Lower* LHD tracts had the highest transition rate before designation, predominantly horizontal transitions to *Hispanic/Lower* and upgrading to *Nonfamily/Moderate*. Although their transition rate decreased during and after designation, upward movements to *Nonfamily/Moderate* neighborhoods were still the most common pattern after designation. In contrast, the transition rates in the other two lower-income types declined during and after designation despite their high retention rate before designation. During and after designations, the transitions from *Hispanic/Lower* neighborhoods were mainly horizontal transitioning to *Nonfamily/Lower* neighborhoods, while upgrading to moderate-income neighborhoods was prevalent in *Black/Lower* neighborhoods. *Nonfamily/Moderate* LHD tracts were much more likely to persist in the same cluster before and during designation, but several downward transitions to *Nonfamily/Lower* neighborhoods were observed after designation.

## References

- Ahlfeldt, G. M., Moeller, K., Waights, S., & Wendland, N. (2017). Game of zones: The political economy of conservation areas. *The Economic Journal*, 127(605), F421–F445. <https://doi.org/10.1111/econj.12454>
- Allison, E., & Peters, L. (2011). *Historic preservation and the livable city*. John Wiley & Sons.
- Arlotta, A., & Avrami, E. (2020). Preservation's engagement in questions of inclusion: A literature review. In E. Avrami (Ed.), *Preservation and social inclusion* (pp. 224–239). Columbia Books on Architecture and the City.
- Been, V., Ellen, I. G., Gedal, M., Glaeser, E., & McCabe, B. J. (2016). Preserving history or restricting development? The heterogeneous effects of historic districts on local housing markets in New York city. *Journal of Urban Economics*, 92, 16–30. <https://doi.org/10.1016/j.jue.2015.12.002>
- Birch, E. L. (2002). Having a longer view on downtown living. *Journal of the American Planning Association*, 68(1), 5–21. <https://doi.org/10.1080/01944360208977188>
- Birch, E. L. (2005). *Who lives downtown*. The Brookings Institution. [https://www.brookings.edu/wp-content/uploads/2016/06/20051115\\_Birch.pdf](https://www.brookings.edu/wp-content/uploads/2016/06/20051115_Birch.pdf)
- Birch, E. L., & Roby, D. (1984). The planner and the preservationist: An uneasy alliance. *Journal of the American Planning Association*, 50(2), 194–207. <https://doi.org/10.1080/01944368408977175>
- Bologna Pavlik, J., & Zhou, Y. (2023). Are historic districts a backdoor for segregation? Yes and no. *Contemporary Economic Policy*, 41(3), 415–434. <https://doi.org/10.1111/coep.12590>
- Buckley, J. M., & Graves, D. (2016). Tangible benefits from intangible resources: Using social and cultural history to plan neighborhood futures. *Journal of the American Planning Association*, 82(2), 152–166. <https://doi.org/10.1080/01944363.2016.1141663>
- Charrad, M., Ghazzali, N., Boiteau, V., & Niknafs, A. (2014). NbClust: An R package for determining the relevant number of clusters in a data set. *Journal of Statistical Software*, 61(6). <https://doi.org/10.18637/jss.v061.i06>
- Cheong, C. (2021). Preservation at a crossroads: The need for equity preservation. *Change Over Time*, 10(1), 66–83. <https://doi.org/10.1353/cot.2021.0002>
- Choi, S., Walter, R. J., & Chalanana, M. (2025). Untangling our checkered past: Investigating the link between local historic district designation and spatial segregation history. *Cities*, 157, Article 105621. <https://doi.org/10.1016/j.cities.2024.105621>
- Cohen, J. R. (1998). Combining historic preservation and income class integration: A case study of the butchers hill neighborhood of Baltimore. *Housing Policy Debate*, 9(3), 663–697. <https://doi.org/10.1080/10511482.1998.9521311>
- Coulson, N. E., & Lahr, M. L. (2005). Gracing the land of Elvis and Beale Street: Historic designation and property values in Memphis. *Real Estate Economics*, 33(3), 487–507. <https://doi.org/10.1111/j.1540-6229.2005.00127.x>
- Coulson, N. E., & Leichenko, R. M. (2004). Historic preservation and neighbourhood change. *Urban Studies*, 41(8), 1587–1600. <https://doi.org/10.1080/0042098042000227028>
- Delmelle, E. (2015). Five decades of neighborhood classifications and their transitions: A comparison of four US cities, 1970–2010. *Applied Geography*, 57, 1–11. <https://doi.org/10.1016/j.apgeog.2014.12.002>
- Fein, D. B. (1985). Historic districts: Preserving city neighborhoods for the privileged. *New York University Law Review*, 60(1), 64–103. <https://heinonline.org/HOL/P?h=hein.journals/nylr60&i=78>
- Florida, R. L. (2002). *The rise of the creative class: And how it's transforming work, leisure, community and everyday life*. Basic Books.
- Foote, N., & Walter, R. (2017). Neighborhood and socioeconomic change in emerging megapolitan nodes: Tracking shifting social geographies in three rapidly growing United States metropolitan areas, 1980–2010. *Urban Geography*, 38(8), 1203–1230. <https://doi.org/10.1080/02723638.2016.1217637>
- Gao, D., & Dublin, J. (2020). Preserving African American places: Growing preservation's potential as a path for equity. *National trust for historic preservation: African American cultural heritage action fund*. <https://savingplaces.org/equity-report>
- Garfinkel-Castro, A., Bushman, T., Sabouri, S., Brewer, S. C., Song, Y., & Park, K. (2020). 6 cluster analysis. In R. Ewing, & K. Park (Eds.), *Advanced quantitative research methods for urban planners* (1st ed.). Routledge. <https://doi.org/10.4324/9780429325038>
- GeoLytics. (2024). Neighborhood change Database 1970–2020. <https://demographics.geolytics.com/ncdb2020/default.aspx>
- GeoLytics (n.d.). Appendix J: Description of tract remapping methodology. <https://geolytics.com/media/pdf/Appendix-J.pdf>
- Gotham, K. F. (2005). Tourism gentrification: The case of New Orleans' Vieux Carré (French Quarter). *Urban Studies*, 42(7), 1099–1121. <https://doi.org/10.1080/00420980500120881>
- Hamer, D. A. (1998). *History in urban places: The historic districts of the United States*. Ohio State University Press.
- Heintzelman, M. D., & Altieri, J. A. (2013). Historic preservation: Preserving value? *The Journal of Real Estate Finance and Economics*, 46(3), 543–563. <https://doi.org/10.1007/s11146-011-9338-8>
- Heuer, T. (2007). Living history: How homeowners in a new local historic district negotiate their legal obligations. *The Yale Law Journal*, 116(4), 768–822. <https://doi.org/10.2307/20455740>
- Hurley, A. (2010). *Beyond preservation: Using public history to revitalize inner cities*. Temple University Press.
- Idziorek, K., & Chalanana, M. (2019). Managing change: Seattle's 21st century urban renaissance. *Journal of Urbanism: International Research on Placemaking and Urban Sustainability*, 12(3), 320–345. <https://doi.org/10.1080/17549175.2019.1598471>
- Kaufman, N. (2013). Putting Intangible Heritage in its place(s): Proposals for policy and practice. *International Journal of Intangible Heritage*, 8, 19–36. <https://doi.org/10.35638/ijih.2013.8.001>
- Kinahan, K. L., & Mawhorter, S. (2021). Tensions between demolition and preservation in Philadelphia. *Journal of Planning Education and Research*, 1–13. <https://doi.org/10.1177/0739456X211042068>

- Kinahan, K. L., & Ruther, M. H. (2020). Uncovering the relationship between historic districts and same-sex households. *Journal of the American Planning Association*, 86(4), 481–494. <https://doi.org/10.1080/01944363.2020.1762507>
- Leichenko, R. M., Coulson, N. E., & Listokin, D. (2001). Historic preservation and residential property values: An analysis of Texas cities. *Urban Studies*, 38(11), 1973–1987. <https://doi.org/10.1080/00420980120080880>
- Listokin, D., Listokin, B., & Lahr, M. (1998). The contributions of historic preservation to housing and economic development. *Housing Policy Debate*, 9(3), 431–478. <https://doi.org/10.1080/10511482.1998.9521303>
- Lovelady, A. (2008). Broadened notions of historic preservation and the role of neighborhood conservation districts. *The Urban Lawyer*, 40(1), 147–183. <https://www.jstor.org/stable/23800827>
- Lubens, R., & Miller, J. (2002). Protecting older neighborhoods through conservation district programs. *Preservation Law Reporter*, 1001–1042.
- Mason, R. (2005). *Economics and historic preservation: A guide and review of the literature*. The Brookings Institution. [https://www.brookings.edu/wp-content/uploads/2016/06/20050926\\_preservation.pdf](https://www.brookings.edu/wp-content/uploads/2016/06/20050926_preservation.pdf)
- McCabe, B. J., & Ellen, I. G. (2016). Does preservation accelerate neighborhood change? Examining the impact of historic preservation in New York city. *Journal of the American Planning Association*, 82(2), 134–146. <https://doi.org/10.1080/01944363.2015.1126195>
- Michael, V. (2014). Diversity in preservation: Rethinking standards and practices. *Forum Journal*, 28(3), 5–12. <https://doi.org/10.1353/fmj.2014.a543386>
- Mikelbank, B. A. (2011). Neighborhood déjà vu: Classification in metropolitan Cleveland, 1970–2000. *Urban Geography*, 32(3), 317–333. <https://doi.org/10.2747/0272-3638.32.3.317>
- Minner, J. (2016). Revealing synergies, tensions, and silences between preservation and planning. *Journal of the American Planning Association*, 82(2), 72–87. <https://doi.org/10.1080/01944363.2016.1147976>
- Morley, J. M. (2006). *Historic preservation & the imagined west: Albuquerque, denver. & Seattle*. University Press of Kansas.
- Napoli, A., Darchen, S., & Spivack, D. (2019). Los Angeles, United States: Adaptive re-use of buildings to regenerate the inner-city. In S. Darchen, & G. Searle (Eds.), *Global planning innovations for urban sustainability* (1st ed., pp. 135–147). Routledge, Taylor & Francis Group.
- National Park Service. (n.d.). How to Apply the National Register Criteria for Evaluation. [https://www.nps.gov/subjects/nationalregister/upload/NRB-15\\_web508.pdf](https://www.nps.gov/subjects/nationalregister/upload/NRB-15_web508.pdf)
- Noonan, D. S. (2007). Finding an impact of preservation policies: Price effects of historic landmarks on attached homes in Chicago, 1990–1999. *Economic Development Quarterly*, 21(1), 17–33. <https://doi.org/10.1177/0891242406296326>
- Noonan, D. S., & Krupka, D. J. (2011). Making-or picking-winners: Evidence of internal and external price effects in historic preservation policies: Evidence of internal and external price effects. *Real Estate Economics*, 39(2), 379–407. <https://doi.org/10.1111/j.1540-6229.2010.00293.x>
- Oba, T., & Noonan, D. S. (2020). The price of preserving neighborhoods: The unequal impacts of historic district designation. *Economic Development Quarterly*, 34(4), 343–355. <https://doi.org/10.1177/0891242420941932>
- Owens, A. (2012). Neighborhoods on the rise: A typology of neighborhoods experiencing socioeconomic ascent. *City & Community*, 11(4), 345–369. <https://doi.org/10.1111/j.1540-6040.2012.01412.x>
- Passell, A. (2021). *Preserving neighborhoods: How urban policy and community strategy shape Baltimore and Brooklyn*. Columbia University Press.
- Reibel, M. (2011). Classification approaches in neighborhood research: Introduction and review. *Urban Geography*, 32(3), 305–316. <https://doi.org/10.2747/0272-3638.32.3.305>
- Rodriguez, M. Z., Comin, C. H., Casanova, D., Bruno, O. M., Amancio, D. R., Costa, L. D. F., & Rodrigues, F. A. (2019). Clustering algorithms: A comparative approach. *PLoS One*, 14(1), Article e0210236. <https://doi.org/10.1371/journal.pone.0210236>
- Rose, C. M. (1981). Preservation and community: New directions in the law of historic preservation. *Stanford Law Review*, 33(3), 473–534. <https://doi.org/10.2307/1228356>
- Ryberg-Webster, S. (2013). Preserving downtown America: Federal rehabilitation tax credits and the transformation of U.S. Cities. *Journal of the American Planning Association*, 79(4), 266–279. <https://doi.org/10.1080/01944363.2014.903749>
- Ryberg-Webster, S. (2017). Beyond rust and rockefeller: Preserving cleveland's african American heritage. *Preservation Education & Research*, 9, 7–23.
- Ryberg-Webster, S. (2019). Community development and historic preservation: Exploring intersections in seattle's chinatown-international district. *Community Development Journal*, 54(2), 290–309. <https://doi.org/10.1093/cdj/bsx046>
- Ryberg-Webster, S., & Kinahan, K. L. (2014). Historic preservation and urban revitalization in the twenty-first century. *Journal of Planning Literature*, 29(2), 119–139. <https://doi.org/10.1177/0885412213510524>
- Rypkema, D. R., Cheong, C., & Mason, R. F. (2013). *Measuring economic Impacts of historic preservation*. Advisory council on historic preservation. <https://www.achp.gov/sites/default/files/guidance/2018-06/Economic%20Impacts%20v5-FINAL.pdf>
- Smith, N. (1979). Toward a theory of gentrification A back to the city movement by capital, not people. *Journal of the American Planning Association*, 45(4), 538–548. <https://doi.org/10.1080/01944367908977002>
- Smith, N. (1998). Comment on David Listokin, Barbara Listokin, and Michael Lahr's "The contributions of historic preservation to housing and economic development": Historic preservation in a neoliberal age. *Housing Policy Debate*, 9(3), 479–485. <https://doi.org/10.1080/10511482.1998.9521304>
- Talen, E., Menozzi, S., & Schaefer, C. (2015). What is a "great neighborhood"? An analysis of APA's top-rated places. *Journal of the American Planning Association*, 81(2), 121–141. <https://doi.org/10.1080/01944363.2015.1067573>
- Wei, F., & Knox, P. L. (2014). Neighborhood change in metropolitan America, 1990 to 2010. *Urban Affairs Review*, 50(4), 459–489. <https://doi.org/10.1177/1078087413501640>
- Wells, J. C. (2021). Does intra-disciplinary historic preservation scholarship address the exigent issues of practice? Exploring the character and impact of preservation knowledge production in relation to critical heritage studies, equity, and social justice. *International Journal of Heritage Studies*, 27(5), 449–469. <https://doi.org/10.1080/13527258.2020.1799059>
- Introduction: Moving past conflicts to foster an evidence-based, human-centric built heritage conservation practice. In Wells, J. C., & Stiefel, B. (Eds.), *Human-centered built environment heritage preservation: Theory and evidence-based practice*, (pp. 1–30). Routledge.
- Werwath, P. (1998). Comment on David Listokin, Barbara Listokin, and Michael Lahr's "The contributions of historic preservation to housing and economic development." *Housing Policy Debate*, 9(3), 487–495. <https://doi.org/10.1080/10511482.1998.9521305>
- Zahirovic-Herbert, V., & Chatterjee, S. (2012). Historic preservation and residential property values: Evidence from quantile regression. *Urban Studies*, 49(2), 369–382. <https://doi.org/10.1177/0042098011404936>
- Zhou, Y. (2021). The political economy of historic districts: The private, the public, and the collective. *Regional Science and Urban Economics*, 86, Article 103583. <https://doi.org/10.1016/j.regsciurbeco.2020.103583>
- Zukin, S. (1987). Gentrification: Culture and capital in the urban core. *Annual Review of Sociology*, 13(1), 129–147. <https://doi.org/10.1146/annurev.so.13.080187.001021>